COURSE TITLE : FABRIC FORMATION I
COURSE CODE : 3062
COURSE CATEGORY : B
PERIODS/WEEK : 4
PERIODS/SEMESTER : 60
CREDITS : 4

TIME SCHEDULE

Module	Topics	Periods
1	Warp & Weft winding	15
2	Warping , Sizing & Drawing-in	15
3	Basic Mechanisms of looms	15
4	Yarn count and Production Calculations	15
TOTAL		60

COURSE OUTCOME

Module	GO	Student will be able to
1	1	Know the warp winding operation in weaving preparatory process
	2	Understand the weft winding operation in weaving preparatory process
2	1	Know the Warping operation in weaving preparatory process
	2	Understand the Sizing operation in weaving preparatory process
	3	Know the Drawing-in operation in weaving preparatory process
3	1	Understand the basic mechanisms in a power loom
	2	Know the shedding motion in a loom
	3	Understand the picking and beating-up motion in a loom
4	1	Understand the yarn numbering system, Heald and Reed count
	2	Know the production and efficiency of weaving preparatory machines

SPECIFIC OUTCOME

MODULE I WARP & WEFT WINDING

1.1.0 Know the warp winding operation in weaving preparatory process

1.1.1 State the objectives of warp winding
1.1.2 List the Functions of Various components of Warp Winding Machine
1.1.3 Illustrate the Working of Electronic Yarn Clearers
1.1.4 Distinguish Electronic Yarn Clearers over Ordinary Slub Catchers
1.1.5 Define the terms in winding
1.1.6 List the different types of knotters and splicers
1.1.7 Explain the working of Roto coner
1.1.8 Describe the working of Auto coner
1.1.9 List the Package Defects in Winding, their causes and remedies

1.2.0 Understand the weft winding operation in weaving preparatory process

1.2.1 State the Objectives of Weft winding
1.2.2 Explain the Working of Schweiter High- speed Automatic Pirn winder
1.2.3 State the Functions of Bunching mechanism and layer locking device
1.2.4 List the defects in pirn winding, their causes and remedies

MODULE II WARPING , SIZING & DRAWING-IN

2.1.0 Know the Warping operation in weaving preparatory process

2.1.1 State the Objectives of Warping
2.1.2 Differentiate between Beam Warping and Sectional Warping
2.1.3 Describe the Working of Ruti high speed Warping machine
2.1.4 Describe the Working of Sectional Warping Machine
2.1.5 State the Defects in Warp beam preparation, their Causes and Remedies

2.2.0 Understand the Sizing operation in weaving preparatory process

2.2.1 State the objectives of Sizing
2.2.2 List the Ingredients used in the Size paste and their Functions
2.2.3 Describe the Working of Multi Cylinder Sizing Machine
2.2.4 State the Functions of the Various Controls in Modern Sizing Machine
2.2.5 List Size recipes for various Cotton yarns
2.2.6 State defects in sized beams their causes and remedies
2.2.8 State the precautions to be taken for Sizing Synthetics and Blends,

2.3.0 Know the Drawing-in operation in weaving preparatory process

2.3.1 State the objectives of Drawing-in
2.3.2 State the objectives of denting
2.3.3 Define the role of Healds and reeds in weaving
2.3.4 Identify the Selection of Healds and Reeds

MODULE III BASIC MECHANISMS OF LOOMS

3.1.0 Understand the basic mechanisms in a power loom

3.1.1 Illustrate the passage of material in a power loom
3.1.2 State the functions of various parts
3.1.3 List the primary, secondary, auxiliary motions in a power loom
3.1.4 State the roles of various motions in weaving

3.2.0 Understand the shedding motion in a loom

3.2.1 State the objectives of shedding
3.2.2 Compare Tappet Shedding, Dobby Shedding and Jacquard Shedding
3.2.3 Illustrate Different types of Sheds
3.2.4 Explain the working of Plain Tappet shedding
3.2.5 State the setting and timing of plain tappet shedding mechanism
3.2.6 State the functions of counter shaft in shedding

3.3.0 Understand the picking and beating-up motion in a loom

3.3.1 State the objectives of picking
3.3.2 Explain the Working of Cone Over pick motion
3.3.3 Describe the Working of under pick motion
3.3.4 Explain the method of Altering Picking force and Setting of Picking band
3.3.5 State the setting and timing of cone over pick mechanism
3.3.6 Explain the beating-up motion in a power loom
3.3.7 Illustrate the Eccentricity of the Sley

MODULE IV YARN COUNT AND PRODUCTION CALCULATIONS

4.1.0 Understand the yarn numbering system, Heald and Reed count

4.1.1 List the different methods of yarn numbering systems
4.1.2 Distinguish between Direct and Indirect Systems of Yarn Numbering
4.1.2 Estimate Equivalent Count
4.1.3 Determine Resultant count, Average count or Beam count
4.1.4 Determine Heald Count and Reed Count Calculations

4.2.0 Know the production and efficiency of weaving preparatory machines

4.2.1 Determine the speed, Production and Efficiency of Warp Winding Machine
4.2.2 Estimate Speed, Production and Efficiency Calculations of Weft Winding Machine
4.2.3 Estimate the Number of Ends per Section & the Number of Sections Required
4.2.4 Determine the production, efficiency and waste in warping machine
4.2.5 Estimate the size percentage
4.2.6 Determine the production and efficiency of sizing machine

CONTENT DETAILS

MODULE I WARP WINDING AND WEFT WINDING

Objectives of Warp Winding- Understand the Working of Warp Winding Machines - Creel - types of tensioners - slub catchers - Broken thread stop motion - Full package stop motion - ribbon breakers - Package arm holders - traversing mechanism - features of Disc and Gate type tensioners - working of Electronic Yarn clearers - Warp Winding -Drum Winding and Precision winding - advantages of Electronic yarn clearers over ordinary slub catchers - Angle of wind - wind ratio - types of knots - splicing - Roto Coner - Auto Coner - defects in winding - causes & remedies

Weft Winding Machines – Objectives - Schweiter high-speed automatic pirn winder - Bunching mechanism - Layer locking device - Pirn diameter control - - defects in pirns, their causes and remedies.

MODULE II WARPING SIZING AND DRAWING

Objectives of warping - beam warping and sectional warping - Ruti high speed warper - functions of various components of a warping machine – Creel - Thread stop motion - Length measuring motion - Head stock - Expanding Comb - doffing motion - Brake Motion - Driving Drum – Tensioners - sectional warping machine -limitations of sectional warping -Defects in warp beams, their causes and remedies

Objectives of sizing - ingredients used in the size paste and their functions - multi cylinder sizing machine - functions of the various controls in a modern a sizing machine-Temperature, Size level –Stretch – Moisture - Viscosity - size receipe for coarse, medium and fine cotton yarns-defects in sized beams - their causes and remedies - precautions to be taken for sizing synthetics and blends - improved fibre lay - after waxing

Objectives of drawing- in and denting – role of Healds and Reeds - Selection of suitable healds and reeds for different fabrics.ODULE I Basic Mechanisms of Looms

MODULE III UNDERSTAND BASIC MECHANISMS IN POWER LOOM

Passage of material through a power loom- various parts and functions – Primary, secondary and auxilary motions and role of each in weaving- Importance of Primary Motions-

Principles of Shedding- Positive and Negative Shedding - Early shedding and late shedding - selection of shedding mechanism- Tappet Shedding- Dobby Shedding- Jacquard Shedding - Different types of Sheds- working of Plain Tappet Shedding mechanism - Setting and timing of plain tappet Shedding – function of counter shaft

Picking- objectives - Positive Picking and Negative Picking- Working of Cone Over Pick motion- Working of Under Pick motions- Method of Altering Picking Force and Setting of Picking band- Timing and Setting of Cone Over Picking - Discuss the beating operation- Explain the working of beating mechanism - Calculate the eccentricity of the sley.

MODULE IV YARN COUNT AND PRODUCTION CALCULATION

Understand the different methods of Yarn Numbering Systems - -direct and indirect systems of yarn numbering - English, French, Metric, Denier, Tex - equations for calculating count in different Direct and

Indirect systems - equations for calculating equivalent count -resultant count of folded yarn with and without contraction - average count (beam count)- heald and reed count .
Speed, production and efficiency of warp winding machines - weft winding machines,warping machines - ends per section and the number of sections required per set - sized yarn weight, un-sized yarn weight and size percentage - production, speed and efficiency of sizing machines.

REFERENCE BOOKS

1. Thomas W.Fox - The Mechanism of Weaving - Universal Publishing corp.,Mumbai -2
2. A.Ormerod - Modern Preparation and Weaving Machinery - Woodhead publishing Ltd., England.
3. Marks & Robinson - Principles of Weaving The Textile Institute, - Manchester.
4. N.N.Banerjee - Weaving Mechanism Vol.I - Smt Tandra banerjee & Smt Apurba banerjee , West Bengal
5. Prof.J.C.Chakravarthi - Mechanism of Weaving Vol.I & II - Prof. J.C.Chakravarthy West Bengal
6. B.Hasmukhrai Fabric Forming - SSMITT&PC, Co-operative stores - Komarapalyam.
7. Woven Fabric Production – I - NCUTE, - New Delhi
8. R.Sengupta - Weaving Calculations - D.B.Taraporevala sons & co Ltd.,